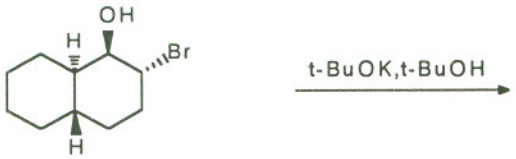
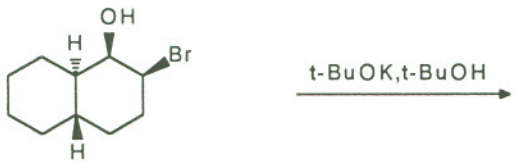
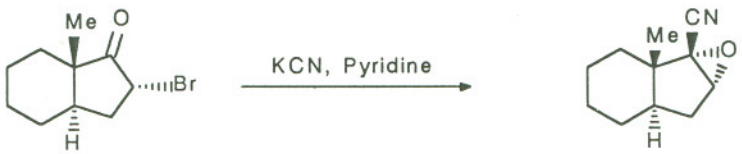
ORGANIC CUME
Oct. 24th 1992

1. The following are a series of simple organic reactions which are heavily influenced by the stereochemistry of the substrate. Using your knowledge of organic chemistry predict the product(s) of these reactions, indicating as much stereochemistry as you can. Where you invoke a stereochemical argument to predict the formation of one stereoisomer rather than another justify your answer. (REMEMBER : A PICTURE SAYS A THOUSAND WORDS) Also indicate whether the transformations are reversible under the reaction conditions, or whether the reaction is irreversible.





2. The following reaction was taken from the literature. Give the mechanism of the reaction and explain the regioselectivity and the stereochemistry of the processes involved during the transformation.



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Reference : Numazawa, M.; Satoh, M.; Satoh, S.;Nagaoka, M.;Osawa, Y. J. Org. Chem. 1986, **51**, 1360.